

Institute of Actuaries of India

Subject CP3-Communication Practice

June 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

Thanks Thomas for writing to me. I am hereby sharing a paper with the necessary material to help you with your Board meeting.

Financial Turbulence

ABC Trust has been involved in providing defined contribution Trust related services to corporate employers. It had been very successful in its endeavour by providing superior return, better services, flexible offerings and portability. As it continues to grow and develop, it should also look to strengthen its offerings against the headwinds.

Recently our economy has faced financial turbulence due to various reasons:

- An increase in interest rates by a large global economy –investors are thus finding better or comparable opportunities overseas
- Depreciation of local currency – global investors are affected as returns reduce due to weaker local currency.
- Default on repayment by large private company – lowers confidence in businesses & system when a high quality company defaults on its promise.

All these factors attributed to lesser interest of investors in local private company issued bonds leading to fall in their market value.

How does it affect ABC Trust

This bond market turbulence has posed a challenge for ABC trust investments. Presently about 25% of trust investments have been made in high quality private corporate bonds. This segment of portfolio has been impacted the most in terms of market value.

A measure of confidence of investors in bond investment is the rating transition matrix. It gives an indication of likelihood of a bond to default on its payments (coupon or redemption). Recent times show a dramatic change in this matrix. For a AA+ bond future expectation to default has increased by more than 15 times. It's a clear indicator that investor confidence is low and they are likely to stay away from new investments in bonds. Hence the prices of these investments are expected stay under pressure for some time in the future.

Such market turbulence presents both risks and opportunities for ABC Trust.

A. Risks for ABC Trust

ABC trust has a liability to honour payments to its client employees when they leave employment.

These payments are uncertain in nature for both amount and timing. Past trends indicate two aspects:

1. To make these payments, ABC trusts first utilize new money it receives as contributions along with income from its investments. In case of any deficit (higher payments, lower contributions or return), there will be a need to sell current investments.
2. Present portfolio has more long term investments and it is likely to overrun expected tenure of payments to honour present investments.

If present financial scenario prevails for long, any tactical or strategic sale of bond investments by trust may realize lower than expected value of investment.

Historically trust had been aggressive in its investment policy and it also helped it gain superior performance (returns) for its investors. Corporate & Government bonds form most portion of portfolio, with the former expected to give higher returns than the latter. A fallout of the current scenario, could be a lower than expected return on investment from portfolio leading to a disappointment by clients.

Downgrades in ratings of bonds indicate market view of quality of investments. Presently market perceives higher future default by other companies. Such defaults may further aggravate the loss of value of investments for trust investment portfolio.

B. Opportunities for ABC Trust

About 5% of the current Trust portfolio is invested in cash or short term money market investments.

This turbulence may also offer some good bargain deals in higher quality & better return investments.

Possible future steps

In light of current scenario, ABC Trust can look to take following steps to review its investment policy:

1. Identify its current exposure to securities which defaulted or where downgrade in rating has happened & evaluate its likely impact on portfolio value.
2. Readjust its asset portfolio to follow future liabilities more closely. This will avoid forced sell situations at inopportune times. An estimate on present structure indicate asset portfolio maturity cash flows are likely to overrun liability cash flows.
3. Develop stronger framework for investment managers in making right investment decisions. A risk\return matrix with increased importance on identification of all risks associated with security will help make investment framework stronger.
4. Enhance disclosure to its customers on current portfolio, risks and investment philosophy.

I hope my thoughts shared above will you in your board meeting. Feel free to connect for any further discussion on this topic

[50 Marks]

Solution 2:

Avoid known information - In preparing response to client inquiry, I notice the audience will be closely involved with the Trust. Hence I decided to exclude information about the trust which would be known to them. E.g. I avoided details provided about historical client details, offerings etc.

Avoid low relevance data - Financial crisis impacted current portfolio & historical performance of fund has no relevance in present scenario. Hence I decided to exclude any use of past data like historical return & CAGR information provided in question.

Add details on complex aspects – E.g. while it seems to be widely known that the financial crisis has hit the market, I have decided to add details on logical implication of its reasons. Similarly, I tried to add a likely impact of asset-liability mismatch on portfolio management.

i)

Dear Ram,

Thank you for your email. I understand the difficulty of communicating the increase in premium to the Client but please be assured that we increase premiums only if the contract does not meet a minimum threshold of profitability. I have outlined below our pricing philosophy and the reasons for the proposed increase.

Our pricing philosophy

In general we expect all contracts to be able to have a minimum margin of 10% to contribute towards our expenses and profitability which means claim payouts cannot be greater than 90% of the premiums. Of course, on a year on year basis we may see some variation which could be purely random or reflective of an underlying

pattern. For large schemes which have been with us for long and for which the claim experience is fairly stable we price with lower margins. For schemes which are new and or are small in order to protect our margins we may allow for higher margins.

In the case of XYZ Company Ltd. when the scheme had been priced originally the scheme had been priced with a higher margin as the client was offering hospitalization benefits to its employees for the first time and therefore there was no group experience on the basis of which we could have priced the scheme. A review of the pricing had been undertaken in CY15 it was concluded that the scheme met the profitability criteria. A more recent review requires us to increase the premium rates. I have given below a summary of the analysis that we have carried out before increasing the premiums.

Profitability of the current contract

You are right that for assessing the profitability we should look at the premiums paid by the Client and compare it with the Claims paid out for the relevant year. However, premiums are paid once a year and cover continues for the next 12 months. This means premium may be paid in a particular financial year however, we could be on cover in the next financial year. Therefore claims pertaining for a particular year of cover will be booked in the next financial year. Another complication in respect of claims is that there is usually a delay in intimating claims to the Company by the customer and therefore we will know the actual claim amount in respect of particular year with a delay. Adjustments are made to the premiums and claims by way of creation of “reserves”, i.e. setting up liabilities so that there is a fair representation of the profitability of the business. For your ease of understanding for the profitability of this contract we have mapped claims to the particular year of cover and shown it in the chart below. This shows that the claims as a percentage of premium have been increasing over the years and profitability of the contract is reducing. I have given below graphical representations of the claim trend.

“Appropriate pictorial representation to enhance understanding”

We further investigated as to the cause of the increase. We are happy to say that there are no underlying profile by age of customers that is causing the claim to increase. However, we do note that across all ages there is an increase in claim and this is due to what is called “Claim Inflation”. Claim inflation is like price inflation and occurs as a result of treatment costs becoming more expensive for certain illnesses. In addition claim inflation occurs as more customers claim health benefits either due to more awareness or due to better diagnosis. We noticed that claims have been inflating by about an average of 4% every year. We are repricing the contract for renewal in CY20. As you are aware these rates are expected to be held for a period of three years. We have therefore increased the premiums so that on an average over the three years the terms are expected to remain in place we have a margin equal to the minimum threshold.

I hope this answers your query. Please do let me know if you have any other questions.

Regards

Shyam

ii)

The material had details of various investigations which had been carried out to conclude on the appropriateness of pricing liked details of experience analysis which I felt were not required to explain the price increase. The material also talked about projecting claims into the future details of which I have excluded from my mail.

iii)

Explanation

I have left out terms like “IBNR”, “unearned premium reserve” and explained “claims inflation” appropriately.

[50 Marks]
